

PROJECT 2 – EXPLORATORY DATA ANALYSIS (EDA)

Dataset: Cleaned_Dataset.xlsx

Problem Statement

The purpose of this project was to explore and analyze the cleaned sales dataset to gain a better understanding of customer purchasing patterns, sales trends, and unusual transactions. The analysis was performed to uncover meaningful insights from the data and support better decision-making.

Methodology

The cleaned dataset obtained from Project 1 was used for this analysis. Descriptive statistics were calculated for numerical columns such as Quantity, UnitPrice, ItemsInCart, and TotalPrice. Different aspects of the dataset, including products, payment methods, order status, and referral sources, were examined to identify trends and patterns. Outliers were detected using the Interquartile Range (IQR) method.

Key Findings

- The dataset consists of 1200 records and 14 columns.
- The average quantity ordered was 2.95 units, while the median quantity was 3 units.
- The average order value was ₹1053.97, and the median order value was ₹823.62.
- Chair was the best-selling product with a total sales quantity of 562 units. Printer and Laptop were the next most popular products.
- Online payment was the most frequently used payment method with 258 transactions.
- Instagram generated the highest number of referrals with 259 records, making it the leading referral source.
- Cancelled orders were slightly higher than the other order statuses recorded in the dataset.
- Outlier analysis identified 8 transactions with unusually high TotalPrice values ranging from ₹3334.00 to ₹3456.40.

Recommendations

- Focus on top-selling products such as Chair, Printer, and Laptop to maintain strong sales performance.
- Continue using Instagram for promotions, as it generated the highest number of referrals.
- Review the identified outlier transactions and regularly monitor sales data to better understand customer behaviour and business trends.